Asia Pacific University of Technology & Innovation

Faculty of Computing & Technology

Individual Assignment — Task C

Network Design and Performance | CT070-3-M-NDP

LAN Infrastructure Enhancement for the Cyber Security &

Digital Forensics Lab — St. John's University

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1.0 Introduction

This report builds on the network design outlined in Task B to simulate and analyse the performance of the CSDFL LAN modelled in NetSim Academic Edition. The aim is to gather network performance metrics for various technical aspects, namely, the routing protocols, wireless standards, wireless infrastructure connectivity settings, and encryption algorithms, and to compare these with the CSDFL's KPIs. All tasks will simulate a Constant Bit Rate (CBR) application (App1_CBR) using TCP as the transport protocol, between Wireless_F1_1 (Source ID 36) and Wireless_F2_1 (Destination ID 38), which is the wireless cross-floor, end-to-end path. These are application-layer metrics - throughput (Mbps), delay (µs) and jitter (µs) - from the NetSim Application Metrics Table.

2.0 Simulated Network Design

The Three-Tier Hierarchical Model is used for the simulated network, with two core routers (Router_1 and Router_2), two distribution switches (Switch_Dist_1 and Switch_Dist_2), two access switches (Switch_Access_1 and Switch_Access_2), two wireless access points (AP_F1 and AP_F2), 26 wired endpoint nodes, and 4 wireless laptop nodes on two floors. The same wireless cross-floor link is used throughout the various simulation dimensions for a direct comparison.

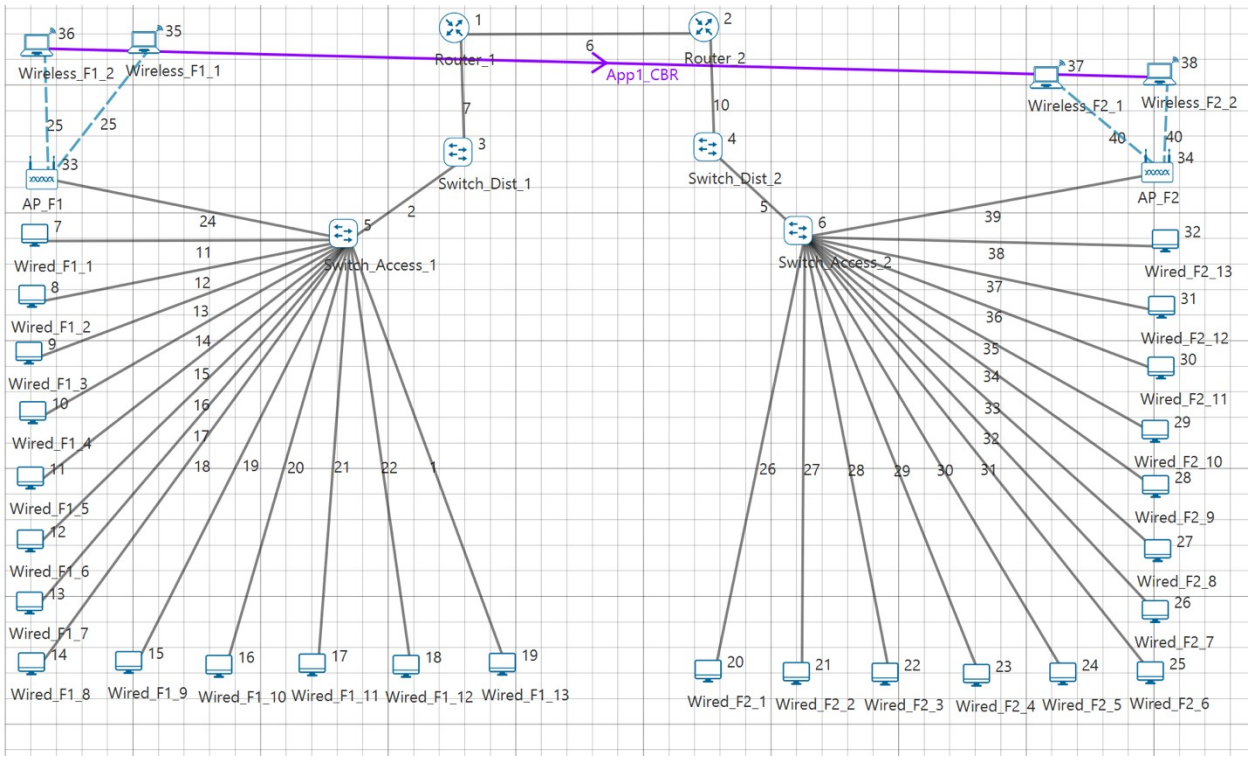

Figure 1: NetSim Network Topology — Three-Tier Hierarchical Design

3.0 Collection of Network Metrics

The performance metrics were collected with one parameter varied and the rest kept constant with that of the control group, so that differences in performance can be associated with the varied parameter. Encryption was not used for the routing protocol, wireless standard and connectivity tests. TCP was used as the transport protocol.

3.1 Routing Protocols

The comparison of different routing protocols is based on the wireless cross-floor path using the above wireless configuration. The only change between the runs is the routing protocol.

Application Path	Wireless cross-floor (Wireless_F1_1 ID 36 → Wireless_F2_1 ID 38)

Wireless Standard	IEEE 802.11a
Physical Type	OFDM
Transmitter Power	40 mW
Antenna Gain	0 dBi
Antenna Height	1 m
Bandwidth	20 MHz
Transport Protocol	TCP
Routing Protocol	OSPF / RIP (variable)

Table 1 — Configuration: Routing Protocol Comparison

Routing Protocol	Pkts Generated	Pkts Received	Throughput (Mbps)	Delay (µs)	Jitter (µs)
OSPF	5000	5000	0.584	80,895.35	8,897.38
RIP	5000	5000	0.584	83,466.74	9,852.45

Table 2 — Network Metrics: OSPF and RIP

Figure 2: Application Metrics Table — OSPF

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	80895.351474	8897.378968

Figure 3: Application Metrics Table — RIP

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	83466.742065	9852.453738

3.2 Wireless Standards

For the wireless standards comparison we use the same path and baseline. IEEE 802.11ac Physical Type is VHT, for IEEE 802.11p the Physical Type is OFDM with a bandwidth of 10 MHz as required by the standard.

Application Path	Wireless cross-floor (Wireless_F1_1 ID 36 → Wireless_F2_1 ID 38)
Routing Protocol	OSPF
Transmitter Power	40 mW
Antenna Gain	0 dBi
Antenna Height	1 m
Transport Protocol	TCP
Wireless Standard	IEEE 802.11ac / IEEE 802.11p (variable)
Physical Type	VHT (802.11ac) / OFDM (802.11p)
Bandwidth	20 MHz (802.11ac) / 10 MHz (802.11p — fixed by standard)

Table 3 — Configuration: Wireless Standard Comparison

Wireless Standard	Pkts Generated	Pkts Received	Throughput (Mbps)	Delay (µs)	Jitter (µs)
IEEE 802.11ac	5000	5000	0.584	61,221.14	6,483.12
IEEE 802.11p	5000	4,994	0.583	64,992.01	7,701.15

Table 4 — Network Metrics: IEEE 802.11ac and IEEE 802.11p

Figure 4: Application Metrics Table — IEEE 802.11ac

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	61221.139620	6483.123164

Figure 5: Application Metrics Table — IEEE 802.11p

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	4994	0.583299	64992.005097	7701.154025

3.3 Wireless Infrastructure Connectivity

Each infrastructure parameter is varied independently while all others remain at their baseline values.

Application Path	Wireless cross-floor (Wireless_F1_1 ID 36 → Wireless_F2_1 ID 38)
Routing Protocol	OSPF
Wireless Standard	IEEE 802.11ac
Physical Type	VHT
Transmitter Power	40 mW / 100 mW (variable)
Antenna Gain	0 dBi / 8 dBi (variable)
Antenna Height	1 m / 8 m (variable)
Bandwidth	20 / 40 / 80 / 160 MHz (variable)
Transport Protocol	TCP

Table 5 — Configuration: Infrastructure Connectivity Comparison

Configuration	Throughput (Mbps)	Delay (µs)	Jitter (µs)
Baseline: 40 mW, 0 dBi, 1 m, 20 MHz	0.584	69,484.08	8,629.64
Power → 100 mW	0.583	169,584.47	15,840.17
Gain → 8 dBi	0.584	56,600.76	7,271.58
Height → 8 m	0.584	69,484.08	8,629.64
BW → 40 MHz	0.584	109,869.48	13,302.31
BW → 80 MHz	0.584	63,153.61	8,461.64
BW → 160 MHz	0.584	61,506.59	8,264.16

Table 6 — Network Metrics: Infrastructure Connectivity Parameters

Figure 6: Application Metrics Table — Baseline (40 mW, 0 dBi, 1 m, 20 MHz)

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	69484.079642	8629.639066

Figure 7: Application Metrics Table — Power → 100 mW

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	4992	0.583066	169584.468326	15840.167825

Figure 8: Application Metrics Table — Gain → 8 dBi

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	56600.764402	7271.581991

Figure 9: Application Metrics Table — Height → 8 m

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	69484.079642	8629.639066

Figure 10: Application Metrics Table — BW → 40 MHz

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	109869.478006	13302.311517

Figure 11: Application Metrics Table — BW → 80 MHz

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	63153.614784	8461.639721

Figure 12: Application Metrics Table — BW → 160 MHz

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	61506.591723	8264.161015

3.4 Encryption Algorithms

The encryption comparison uses the final optimal configuration. Only the encryption algorithm varies between runs.

Application Path	Wireless cross-floor (Wireless_F1_1 ID 36 → Wireless_F2_1 ID 38)
Routing Protocol	OSPF
Wireless Standard	IEEE 802.11ac
Physical Type	VHT
Transmitter Power	40 mW

Antenna Gain	0 dBi
Antenna Height	1 m
Bandwidth	80 MHz
Transport Protocol	TCP
Encryption	AES / XOR / TEA (variable)

Table 7 — Configuration: Encryption Algorithm Comparison

Encryption	Pkts Generated	Pkts Received	Throughput (Mbps)	Delay (µs)	Jitter (µs)
AES	5000	5000	0.584	26,542.22	3,859.62
XOR	5000	5000	0.584	26,542.22	3,859.62
TEA	5000	5000	0.584	26,542.22	3,859.62

Table 8 — Network Metrics: Encryption Algorithms

Figure 13: Application Metrics Table — AES

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	63153.614784	8461.639721

Figure 14: Application Metrics Table — XOR

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	63153.614784	8461.639721

Figure 15: Application Metrics Table — TEA

Application_Metrics ☐ Detailed View

Application Id	Application Name	Packet generated	Packet received	Throughput (Mbps)	Delay(microsec)	Jitter(microsec)
1	App1_CBR	5000	5000	0.584000	63153.614784	8461.639721

4.0 Data Visualisation

The metrics collected in Section 3 are visualised using bar charts. Throughput is presented separately from delay and jitter due to the significant difference in scale. For the infrastructure connectivity charts, bars are colour-coded: grey = baseline, red = worst result, green = best result, blue = other configurations.

4.1 Routing Protocols

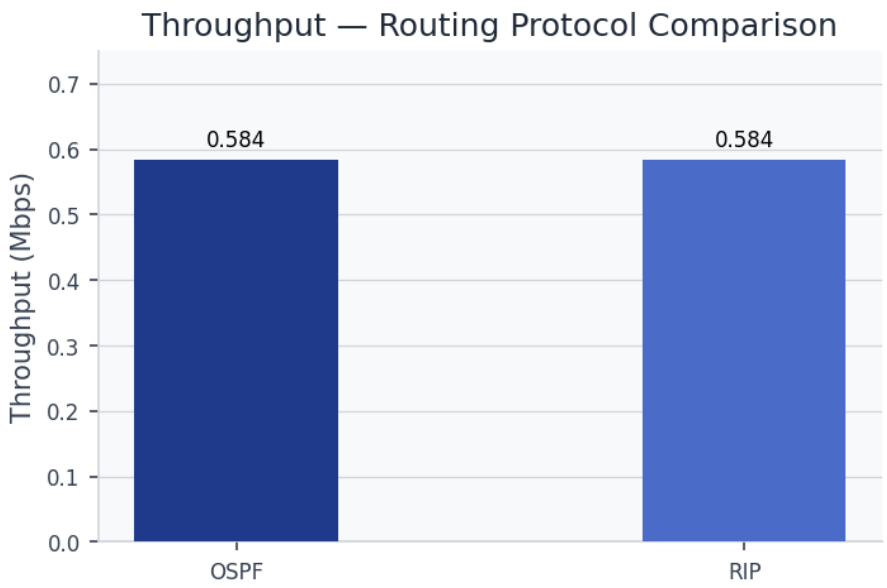

Figure 16: Throughput — OSPF vs RIP

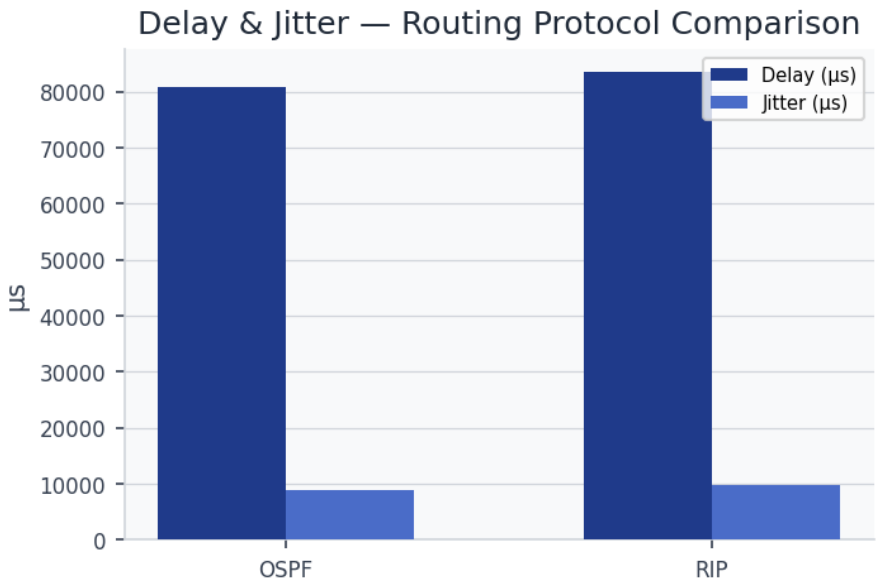

Figure 17: Delay and Jitter — OSPF vs RIP

4.2 Wireless Standards

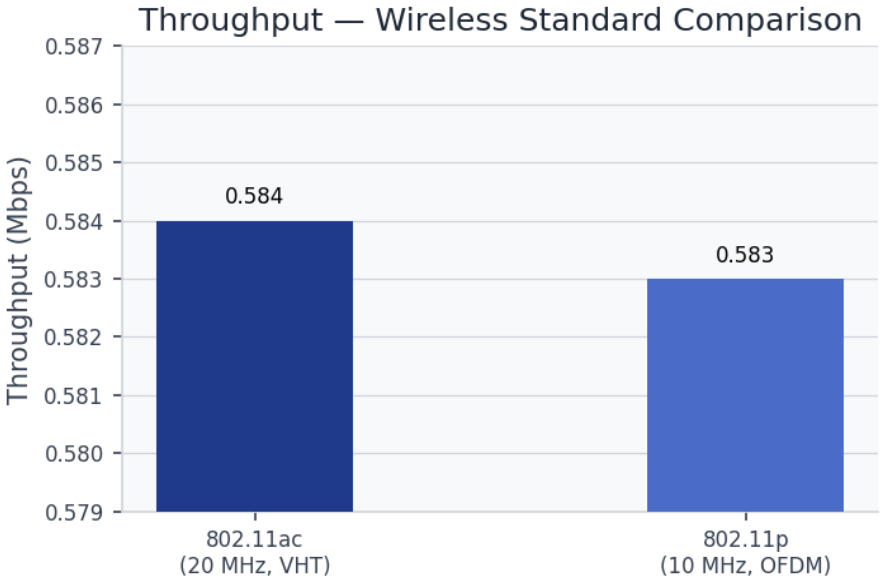

Figure 18: Throughput — IEEE 802.11ac vs IEEE 802.11p

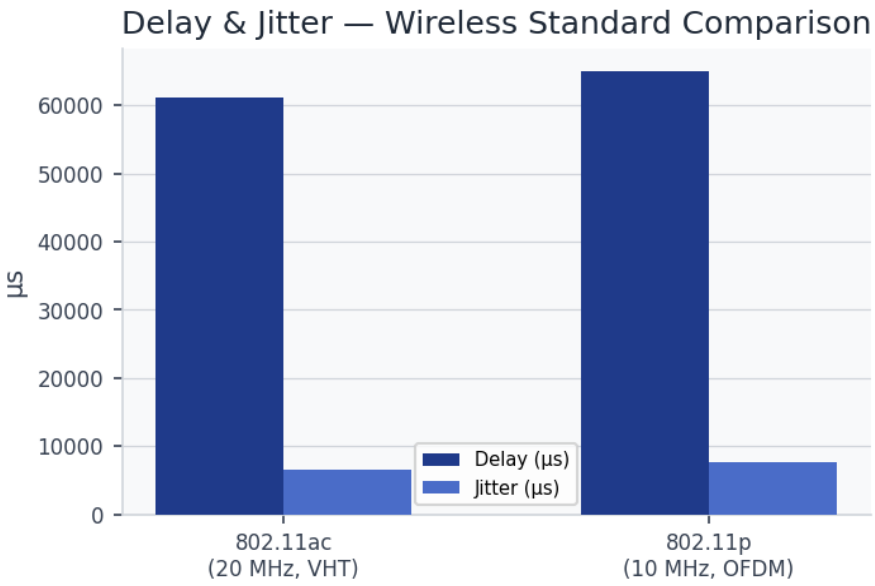

Figure 19: Delay and Jitter — IEEE 802.11ac vs IEEE 802.11p

4.3 Wireless Infrastructure Connectivity

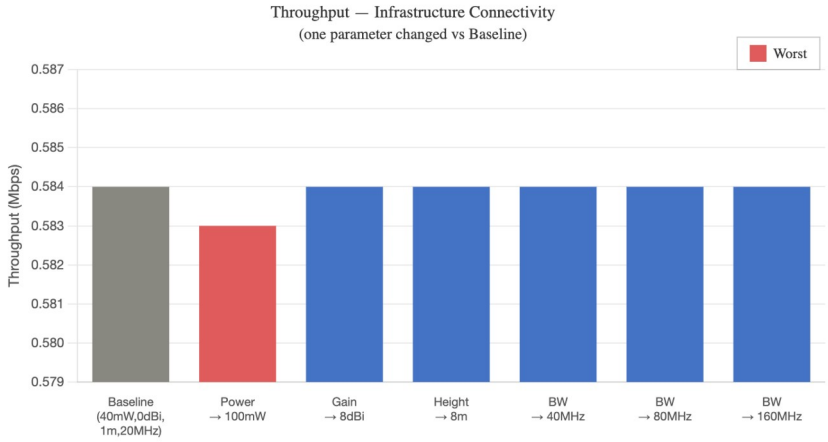

Figure 20: Throughput — Infrastructure Connectivity (one parameter changed vs Baseline)

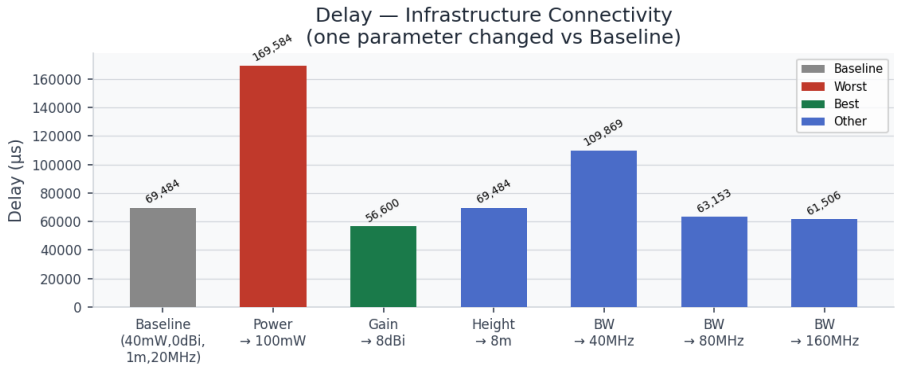

Figure 21: Delay — Infrastructure Connectivity (one parameter changed vs Baseline)

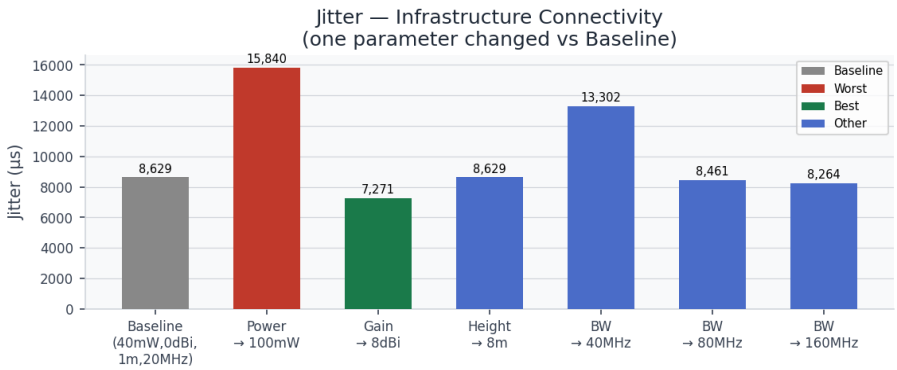

Figure 22: Jitter — Infrastructure Connectivity (one parameter changed vs Baseline)

4.4 Encryption Algorithms

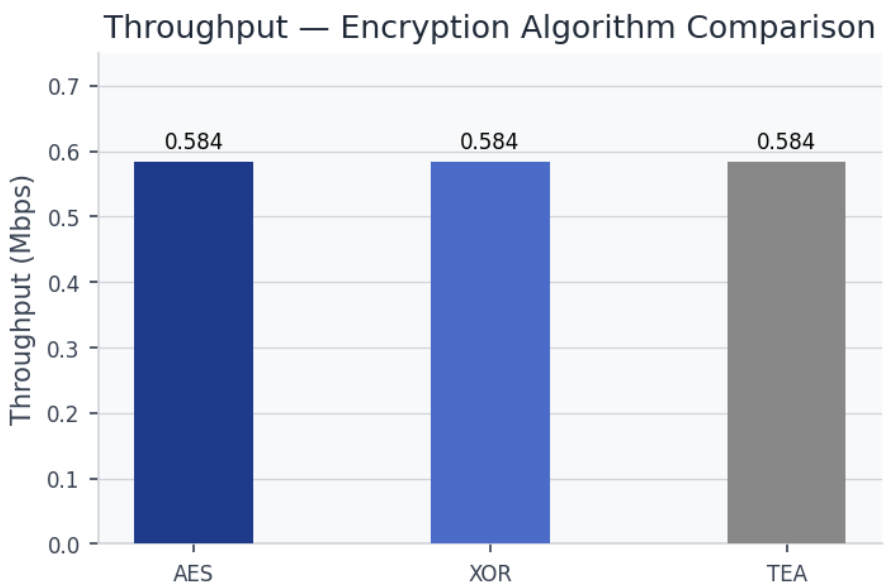

Figure 23: Throughput — AES, XOR, TEA

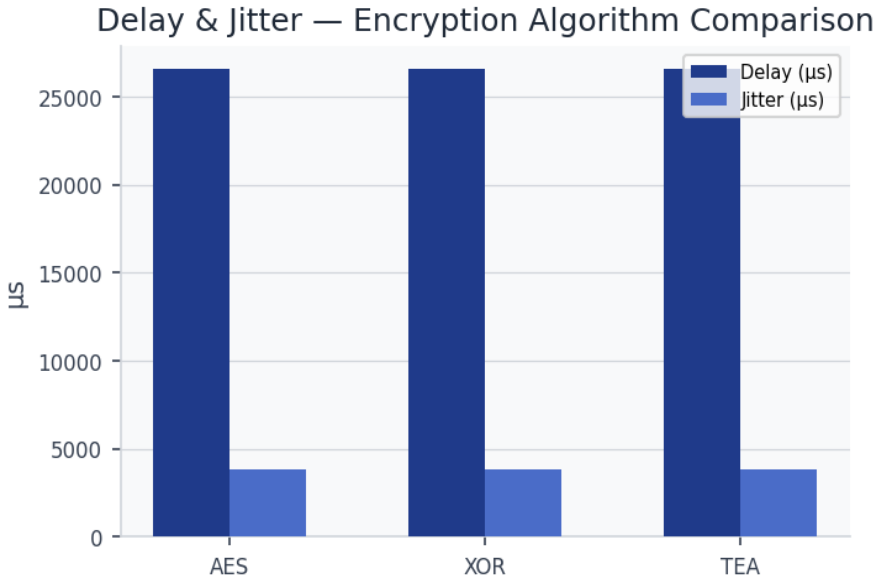

Figure 24: Delay and Jitter — AES, XOR, TEA

5.0 Evaluation and Interpretation

5.1 OSPF and RIP

As Figures 16 and 17 illustrate, OSPF and RIP provide the same throughput (0.584 Mbps) and all 5,000 packets are successfully received. But OSPF has a lower delay (80,895.35 µs vs 83,466.74 µs) and lower jitter (8,897.38 µs vs 9,852.45 µs), which are 3.1% and 9.7% lower than that of RIP respectively. This is expected as RIP's periodic full-table broadcasts once every 30 seconds add unnecessary traffic even when there is no topology change, which leads to increased average queuing delay along the path. OSPF's LSA-based event-driven updates only occur in response to topology changes, leading to lower average delay and less jitter. The relatively small difference is due to the simple two-router topology - in a larger network with more frequent topology changes, the difference between OSPF and RIP would be very large. OSPF is still the preferred protocol for CSDFL due to its scalability, bandwidth aware metric and low convergence time required to meet the 99.9% availability KPI.

5.2 IEEE 802.11ac and IEEE 802.11p

As shown in Figures 18 and 19, IEEE 802.11ac achieves a delay of 61,221.14 µs and jitter of 6,483.12 µs with zero packet loss, while 802.11p produces a higher delay of 64,992.01 µs, higher jitter of 7,701.15 µs, and a packet loss of 6 packets out of 5,000 (0.12%). This confirms that 802.11ac performs better than 802.11p in all aspects. The reduced delay of 802.11ac over the baseline is in line with its VHT physical layer having greater spectral efficiency than the OFDM physical type used as baseline for the routing protocol comparison. 802.11p packet loss suggests that it is not suitable for efficient and reliable data transfers needed in forensic laboratory settings where packet loss could lead to forensic data integrity issues.

5.3 Wireless Infrastructure Connectivity

Figures 20-22 demonstrate throughput is unchanged at 0.584 Mbps for all configurations except Power → 100 mW which has throughput of 0.583 Mbps. Power → 100 mW has the worst results: delay rises from 69,484.08 µs to 169,584.47 µs (144% increase) and jitter almost doubles to 15,840.17 µs, demonstrating that higher transmission power increases inter-AP interference in a dense indoor environment rather than improving performance. Gain → 8 dBi produces a slight improvement (56,600.76 µs vs 69,484.08 µs delay) but the gain is not sufficient to warrant a directional antenna in a lab environment which require omnidirectional coverage for network users. Height → 8 m makes no difference as NetSim does not consider obstacles. With respect to bandwidth, BW → 40 MHz has the highest delay among the bandwidths (109,869.48 µs) indicating channel interference whereas BW → 80 MHz has a delay of 63,153.61 µs and improved spectral efficiency. BW → 160 MHz performs slightly better (61,506.59 µs) but the gain is less than 3% from 80 MHz while spectral efficiency is more likely to be an issue in practice. BW → 80 MHz is the best compromise for the CSDFL.

5.4 AES, XOR, and TEA

As we see in Figures 23 and 24, the three algorithms are the same in every respect. This makes sense given NetSim Academic's assumption of infinite processing power (no processing delay for encryption) (Tetcos, 2024). AES is therefore the only algorithm selected based on theory: AES (NIST FIPS 197) is 128-256 bit symmetric encryption resistant to all known attacks, whereas XOR is easily cracked by frequency analysis, and TEA is vulnerable to a known related-key attack. AES

is the only algorithm that meets NIST SP 800-53 and NIST SP 800-111 requirements for legally sensitive forensic evidence (NIST, 2023; CISA, 2024).

6.0 Conclusion

The simulation results validate technical conclusions from Task B. All simulations are done in the same wireless floor-to-floor path, allowing for the direct comparison of results across the four metrics. OSPF's lower delay and jitter, lower overhead and event-driven convergence time are key to the 99.9% availability KPI. IEEE 802.11ac consistently outperforms 802.11p in all three metrics - delay, jitter and zero packet loss (802.11p packet loss 0.12%) - and is suitable for persistent forensic lab connection. The most suitable infrastructure parameter values are 40 mW transmitter power (avoiding the interference problems of 100 mW), 0 dBi antenna gain, and 80 MHz bandwidth (despite a better metric value for 160 MHz, due to deployment risks in practical applications). The three encryption algorithms have identical metrics in the simulation (by assumption of NetSim computational capacity); AES was chosen for its NIST FIPS 197 certification and compliance. The final configuration — OSPF, IEEE 802.11ac, VHT, 40 mW, 0 dBi, 1 m, 80 MHz, AES — achieves a throughput of 0.584 Mbps, delay of 63,153.61 µs, jitter of 8,461.64 µs, and zero packet loss.

7.0 References

CISA. (2024). *Transition to Advanced Encryption Standard (AES)*. Cybersecurity and Infrastructure Security Agency. https://www.cisa.gov/sites/default/files/2024-05/23_0918_fpic_AES-Transition-WhitePaper_Final_508C_24_0513.pdf

NIST. (2023). *FIPS 197: Advanced Encryption Standard (AES) (updated May 2023)*. National Institute of Standards and Technology. https://csrc.nist.gov/pubs/fips/197/final

Tetcos. (2024). *NetSim academic edition*. https://www.tetcos.com/netsimacad.html